

COOLING SLEEPING PAD USING PELTIER MODULE (TEC1-12706)

Sonia Azhar Siddiqui - Amir Rehman - Abdul Ghani - Arham Jaffer

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Components and Roles:

1. **Peltier Module (TT271, 12V, 10A):** Acts as the primary cooling element. One side gets cold while the other heats up when powered.
2. **Mini Metal Tank:** To hold water that will be pumped through the cooling system.
3. **Water Pump:** Circulates water from the tank through the pipes across the sleeping pad.
4. **Insulating Foil:** To reduce heat loss and maintain cooling efficiency in certain areas.
5. **Thermal Paste:** Ensures efficient heat transfer between the Peltier module and the heat sink.
6. **High Index Polymeric Pad:** This pad will spread the cooling effect evenly across the surface of the sleeping pad.
7. **Copper Pipe:** To circulate water through the system; copper is used for high conducting efficiency.
8. **Thin Plastic Drip Pipes:** For connecting two copper pipes at the bends.
9. **Thermal Sheet:** To spread cold across the surface of the pad for even cooling.
10. **Regulator:** Controls the temperature by adjusting the power going to the Peltier module, maintaining comfort.
11. **Thermometer:** Measuring temperature for experimental analysis.
12. **Heat Sink (Fan):** To dissipate the heat from the hot side of the Peltier module.
13. **Padding:** Fabric layer for comfort.

Steps to Build:

1. Create Water Circulation System

- **Tank Setup:** Fill the mini tank with water. This tank will be the water container.
- **Pipes:** Connect the water pump to the mini tank with one end of a plastic thin pipe. The pump will push water through the pipe into the cooling pad system.
- **Cooling Pad Setup:** Lay the high index polymeric pad across your sleeping area. Embed the thin copper pipes, connected at the bends with plastic pipes, within the

pad to circulate the water evenly. Ensure the pipe runs throughout the pad to distribute cooling uniformly.

- **Return Pipe:** Connect the plastic return pipe back to the tank to create a closed-loop system for water circulation.

2. Peltier Cooling Setup

- **Attach Peltier Module:** Attach the cold side of the Peltier module to the mini tank using thermal paste for efficient cooling of the water. You can also attach a thermal sheet on the cold side to improve heat transfer.
- **Heat Dissipation:** Attach the heat sink to the hot side of the Peltier module. Use thermal paste to improve heat transfer and attach a fan to the heat sink to blow away the heat and maintain cooling efficiency.
- **Insulate the Tank:** Wrap the tank in insulating foil to reduce external heat loss, ensuring that the cooling effect is retained in the water.

3. Temperature Control

- **Regulator:** Connect the thermostat to monitor the water or pad temperature. Use the regulator to control the power going to the Peltier module, regulating the cooling based on the desired sleeping pad temperature.

4. Assembly and Integration

- Ensure that all components are properly connected:
 - Water pump connected to the tank.
 - Pipes embedded in the sleeping pad.
 - Peltier module cooling the water in the mini tank.
 - Heat sink and fan dissipating the heat from the Peltier.
- Secure everything to make it portable, leak-proof, and functional as a cooling system.

How It Works:

- **Cooling Cycle:** Water in the tank is cooled by the Peltier module. The water pump circulates this cooled water through the pipes embedded in the sleeping pad, keeping the surface cool.
- **Heat Dissipation:** The heat generated on the hot side of the Peltier module is dissipated by the heat sink and fan, ensuring the cooling side remains efficient.
- **Temperature Regulation:** The regulator controls the power going to the Peltier, allowing you to maintain a comfortable temperature.
- **Padding:** Cover the system with an additional layer of fabric to improve comfort.

Diagram:

